Ritwick Verma

Backend Engineer · C++ · Search Systems
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SUMMARY

Backend engineer with 5.5 years on Cassini, the distributed C++ search engine behind every ebay.com search (1B+ listings). Owned the fitment subsystem and built caching, query-rewriting and embedding plumbing on its serving path. Joined as SE2 (university hire), promoted to SE3; 215 merged PRs, 13th of 121 contributors to the engine.

EXPERIENCE

eBay

Software Engineer 3, Search Backend (promoted), San Francisco, CA — Jun 2023 to Present

- Owned eBay Motors fitment, the C++ subsystem that decides which parts fit a buyer's vehicle, across 16 sites and 19 catalogs; top author of its current code (31% of lines). Designed the query operator clients use to request fit constraints or a per-item "fits" signal.
- Built a request-scoped result cache that removed repeat backend calls across recall-recovery passes. Added client-supplied embeddings and batching under the embedding service's limits, and fixed a dead embedding cache.
- Built three C++ query transforms for the product-grouped carousels on the results page, checked against replayed traffic. Folded three ranking expressions into one native C++ macro that matched production on live items.
- Moved fitment's production index and the engine's test fixtures off a legacy query language with no flag day, keeping 2,549 unit tests and 511 regression cases green, after auditing 404 deployed expressions across 4,815 configs.
- Before merging a fitment refactor, replayed 50,000 production-sampled queries, traced 160 new failures to an init-order dependency in shared config and fixed the root cause (new errors 167 to 6). Fixed a production memory spike by capping a combinatorial expansion in query rewriting.
- Made the tokenizer stack allocator-aware (C++ PMR) in 7 PRs and shipped it as an internal HTTP API that science and QA teams use to debug tokenization.
- Wrote the markdown-based test frameworks the team adopted as its standard and a language server for the engine's query language (719 operators), and added cross-environment result diffing to the team's query CLI.

Software Engineer 2, Search Backend, San Jose, CA — Mar 2021 to Jun 2023

- Rewrote fitment for garage vehicles and tire and wheel specifications (57 fitment PRs in 2022), with spec translation for the German, French and Italian sites, and added A/B testing that runs control and treatment in the same request.
- Reworked the engine's core query, table and property containers to use C++ polymorphic allocators.

Earlier: software intern at Shoelace Wireless (Android, 2020) and BHEL (Java EE, 2017).

SELECTED PROJECTS

Side projects are written with AI coding agents, mostly Hermes, an open-source agent I run on my own hardware. I make the design calls and check the results. Also ShutUp, an Android app with 26,000+ downloads (v1 written by hand in 2016).

- **TizenAmbilight:** Ambilight from inside a Samsung TV. The public capture API only returns the UI layer, so it reads the undocumented video plane instead, tone-maps HDR and drives an LED strip at 20 to 25 fps.
- **Crush 80 firmware:** Maintain a fork of a community ZMK port for a keyboard with no vendor source: added a 154-LED per-key RGB driver over SPI-DMA, OpenRGB support and one-cable install.
- **Plutus:** Ran 30+ trading-strategy experiments on point-in-time data with bootstrap confidence intervals. None showed an edge that held up, and the write-up says so.
- **Homelab as code:** A hobby: 16 Proxmox guests, 14 declared in OpenTofu, with a nightly CI job that flags config drift.

SKILLS

- **Languages:** C++ (17/20/23), Python, Java, C
- **C++ tooling:** Polymorphic allocators (PMR), ASan, clang-tidy, CMake, GoogleTest
- **Search:** Query rewriting, tokenization, recall and ranking factors, embedding retrieval, caching, A/B tests
- **AI tools:** Claude Code, agent tooling and MCP servers
- **Tools:** Linux, Docker, Git, GitLab CI

EDUCATION

Master of Computer Science, University of California, Irvine — 2019 to 2020
B.Tech, Computer Science and Engineering, SRM Institute of Science and Technology — 2015 to 2019